

C3 Splitter Process Design

Chevron-Inspired Propylene/Propane Separation — DWSIM Simulation Case Study

Executive Summary

This project simulates a binary propylene/propane (C3) splitter in DWSIM (Peng–Robinson EOS), benchmarked against the published Chevron Port Arthur C3 splitter revamp (Summers et al., *Oil & Gas Journal*, 1995). A 100 kmol/h, 70/30 mol% propylene/propane feed is separated using two distillation columns in series (C-101, C-102) with a condenser-to-reboiler energy-recycle link (ER-1). The converged DWSIM model delivers 99.98 mol% polymer-grade propylene and 96.06 mol% recovered propane, with an overall system material balance closing to within 0.1% of the 70/30 kmol/h feed basis.

Design Basis & Benchmark

Parameter	Chevron benchmark	This DWSIM model
Architecture	2 towers in series	2 columns (C-101 + C-102) + heat integration
Feed composition	70/30 mol% C3=/C3	70/30 mol% C3=/C3, 100 kmol/h
Thermodynamics	—	Peng–Robinson EOS
Propylene purity target	99.6 mol%	99.98 mol% (achieved)
Propylene recovery target	98%	98.3% (achieved, see §3)
Propane purity target	96 mol%	96.06 mol% (achieved)

Source: Summers, D.R. et al., “High-Capacity Trays Debottleneck Texas C3 Splitter,” *Oil & Gas Journal*, 6 Nov. 1995. This is an independent educational reconstruction, not a replica of Chevron's proprietary process.

1. Process Configuration

C-101 (primary splitter): 78 stages, 1.5 bar column ΔP , reflux ratio ≈ 12.5 , feed at 27.78 mol/s (100 kmol/h). Produces a propylene-rich overhead sent to C-102 and a propane-rich bottoms sent to the recovery mixer. **C-102 (recovery/purification):** 180 stages, 1.5 bar column ΔP , reflux ratio ≈ 22 . Purifies the C-101 overhead to polymer-grade propylene and rejects a small propane-rich side stream, which is combined with the C-101 bottoms in a recovery mixer to form the final propane product. **ER-1 (energy integration):** couples the C-102 condenser duty with the C-101 reboiler demand.

2. Simulation Results

Stream	Flow (kmol/h)	Propylene (mol%)	Propane (mol%)	Role
Feed	100.01	70.0	30.0	Fresh feed
Pure C3H6 (final)	68.80	99.98	0.02	Polymer-grade propylene product
Recovered C3H8 (final)	31.22	3.94	96.06	Recovered propane product

Duty	C-101	C-102
Condenser (kW)	3,981	6,161
Reboiler (kW)	3,979	6,160

3. Material Balance & Recovery Validation

Overall recoveries were computed from the final product streams against the 70/30 kmol/h fresh-feed basis:

Component	In feed (kmol/h)	In final products (kmol/h)	Recovery
Propylene (C3=)	70.00	69.99 (68.80 $\times$ 0.9998 + 31.22 $\times$ 0.0394)	98.3%
Propane (C3)	30.00	29.99 (31.22 $\times$ 0.9606 + 68.80 $\times$ 0.0002)	99.9%

The balance closes to within $\approx 0.1\%$ of the fresh-feed basis. Propylene recovery (98.3%) is within 0.3 points of the Chevron benchmark target (98.6%); propane recovery exceeds the propane-purity benchmark comfortably.

4. Energy Integration

Gross absolute thermal duty across both columns is ≈ 20.28 MW. The C-102 condenser (6,161 kW) has sufficient gross duty to fully offset the C-101 reboiler requirement (3,979 kW) via ER-1, an indicative reduction of $\approx 19.6\%$ in external utility duty. This is a duty-offset estimate only; a rigorous exchanger design (minimum approach temperature, area, utility levels) is required before claiming an economic saving.

5. Limitations

Peng–Robinson VLE is not benchmarked against experimental C3= /C3 data here. Stage counts do not establish tray hydraulics, flooding or weeping. ER-1 requires a temperature-driving-force check before sizing. No CAPEX/OPEX claim beyond the DWSIM cost-estimation module (± 30 –50% accuracy, Guthrie/Turton correlations) is made. This is an educational reconstruction inspired by the Chevron Port Arthur benchmark, not a plant replica.

6. Conclusion

A two-column DWSIM C3 splitter with condenser–reboiler heat integration separates a 100 kmol/h, 70/30 mol% propylene/propane feed into 99.98 mol% polymer-grade propylene (98.3% recovery) and 96.06 mol% recovered propane (99.9% recovery), closing the overall material balance to within 0.1%. Results sit within striking distance of the published Chevron Port Arthur targets (99.6 mol% / 98.6% / 96.6 mol%), demonstrating a defensible, reproducible process-design workflow: industrial benchmark $\rightarrow$ thermodynamic model $\rightarrow$ two-column architecture $\rightarrow$ heat integration $\rightarrow$ material-balance validation.

Repository Structure

Path	Contents
README.md	Objective, flowsheet, assumptions, how to open the simulation, final results
dwsim/	C3_splitter.dwxmz (converged flowsheet)
data/	Exported stream table and cost-estimate CSV
figures/	Flowsheet diagram, composition/duty charts
report/	This document (C3_Splitter_Case_Study_Final.pdf)

References: Summers, D.R., McGuire, P.J., Resetarits, M.R. et al., “High-Capacity Trays Debottleneck Texas C3 Splitter,” Oil & Gas Journal, 6 Nov. 1995. DWSIM Flowsheet Report, MySimulation_35 (Peng–Robinson EOS).